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1/19/1982

Thank you John.

Our next new product is the star of today's show, the TRS-80 Model 16. ** 1** As you might surmise from it's model number it is a 16 bit computer and is our new top of the line computer system. This newest TRS-80 continues our tradition of offering the best price/performance in the industry as well as expanding our product line upward to a whole new group of potential customers.

Externally you can see that the Model 16 has an stylish white case and ** 2** a green screen high resolution monitor. To the right of the screen are mounted one or two all new thinline eight inch double sided disk drives. The keyboard ** 3** is a separate movable piece and its layout is the same as the the TRS-80 Model II. In addition to its good looks and visible external features the Model 16 has the internal power that make it today's most powerful desktop computer.

Lets examine these internal features in a little more detail. ** 4** First of all the Model 16 features dual

microprocessors, the MC68000 16/32 bit processor and the Z80A 8 bit processor. This use of dual processors both enhances operation and adds a new dimension in compatibility. ** 5** The internal RAM memory capacity is a half a megabyte, 512K. This memory starts at 128K and can be expanded in 128K steps. The MC68000 is capable of addressing much more memory but the physical limitation of the Model 16 is 512K. ** 6** The disk drives are double sided, double density 8 inch drives with a capacity of 1.25 megabytes each for a total capacity of 2.5 megabytes inside the case. Two additional floppy drives may also be added externally for a total capacity of 5 megabytes.

** 7** A high resolution 12 inch monitor with a green screen is used to enhance readability and reduce glare. A high resolution graphics option as we announced for the Model II will also be made available for the Model 16. ** 8** The use of the dual processors allows full Model II software compatibility. All current Model II software will operate on the Model 16 in an 8 bit Model II mode. This means that all of our software plus hundreds of third party programs are already available that can operate on the Model 16.

**** 9**** A basic configuration Model 16 looks like this. It would have a single disk drive and 128K RAM memory, with two serial communications ports one of which is capable of bisynchronous communications to IBM and other mainframes. The price of this very powerful desktop micro ****10**** is only \$4999. We feel that this is a very competitive price for a true 16 bit computer.

****11**** The basic system can be expanded in increments as shown. An additional disk drive is \$799. A two drive, 128K Model 16, would thus cost \$5798 and would be capable of running large diskette programs without the need for additional external disk drives. The memory boards in the Model 16 hold 256K of RAM so the second 128K is just a set of RAM IC's for \$499. The third 128K is another memory board for \$699, and the last 128K another set of RAMs for \$499 to bring the system to a full 512K.

****12**** A fully configured Model 16 is \$7495. In this configuration it would have 8 times the memory of a Model II and 25% more disk capacity than a Model II with 4 disk drives yet it is still all contained in one compact desktop case.

****13**** We feel one the most popular configurations of the Model 16 will be with our new hard disk drive. For just under \$10,000 this system would provide 256K of RAM memory, 8.4 megabytes of hard disk storage and a single 1.25 megabyte floppy disk for backing up the hard disk drive.

****14**** This system, with the addition of a printer and software, would be capable of storing all the programs and data required to do all the accounting of a relatively large small business. It would offer high speed processing, due to both the 16 bit processor and the hard disk, that could not be matched by a floppy disk system. We started shipping the hard disk for use with the Model II in early December and it has already proven to be a hot seller as well as a very reliable product.

Initial deliveries of the Model 16 will start in about six weeks. It will be capable of being used as a floppy disk based system or a hard disk system from the first as our hard disk is used with the Model 16 without modification.

****15**** This chart will give you an idea of where we are positioned compared to our competition. It is hard to make such a chart an exact comparison as some of the competitors charge extra for items that are included with

our products or do not offer the same communications flexibility. Still I think it is obvious that we offer a very strong price/performance advantage compared to all the other popular small business systems.

****16**** A more graphic way of showing the price performance is to show the computer cost in relation to disk storage capacity. On this graph the higher the bar the higher the cost of that capacity. The Model 16 is clearly the leader. Disk capacity is critical to the small business use of a computer for accounting or inventory control. We have learned that the use of 5 1/4 inch disk drives will restrict using a micro for small business accounting to very small companies. In addition a 5 1/4 inch drive is not adequate as backup device for any but the lowest capacity hard disk.

A 16 bit microprocessor provides two major advantages over an 8 bit processor. First of all it can directly address several megabytes of memory while an 8 bit is limited to 64K. Those micros that address more than 64K with an 8 bit processor sacrifice operating speed and generally have software problems. This ability to directly address large core memory is of significant importance to such memory intensive applications as financial planning,

electronic filing and word processing. The other advantage of a 16 bit processor is that it is faster. Instead of dealing with 8 bit size internal data registers it can deal with 16 bit size registers and, in the case of our MC68000, in 32 bit registers. While the speed of the processor is rather invisible to the end user in many applications there is one important application of a 16 bit system that takes advantage of both the speed and the large memory.

That application is called multitasking and it consists of having the computer run 2 or more programs in its memory at the same time. ****17**** This slide illustrates a typical application of multitasking where two dumb terminals are connected to a Model 16 by the serial ports. Each user can run a totally independent application simultaneously. The Model 16 is hardware equipped to support three simultaneous users, one at it's keyboard and two at the terminals. This feature gives the user a powerful expansion option that he can implement whenever he wants. In this manner the Model 16 can grow to the equivalent of three separate computers at a very low incremental cost to meet a growing user's needs.

****18**** The operating system for the Model 16 has been designed and implemented by us. In it's first release

version it will not support multitasking but we plan a second release by June end with multitasking in place. Operating systems have been getting a lot of press lately with predictions of various systems becoming the 16 bit standard. Our operating system is as close as possible to the Model II system which has had excellent user acceptance. The commands and syntax are identical allowing any Model II user to feel immediately at home with a Model 16. Our operating systems are all designed to meet the needs of the first time computer user and offer a far better user interface, more security from data loss and better documentation than any of the so called standards.

To aid outside software development on the Model 16 we have included a full assembly language development system including an editor, macro assembler, linking loader, cross reference and debugger. A single command will cause the system to go into a Model II mode and the user can simply insert a Model II diskette to operate in the Model II mode.

****19**** The same COBOL that is available for the Model II and III computers will be available for the Model 16. This will allow moving of many of our applications to the Model 16 as well as allowing many minicomputer applications packages to moved to the 16 with little effort.

During the year we will add a BASIC interpreter and compiler plus FORTRAN.

****20**** One of the most important reasons for our dual processor design was to provide a backward compatibility that is virtually unique in our industry. Our design has allowed us to build a Model 16 upgrade kit that will transform any Model II into a Model 16 for only \$1499.

****21**** This kit consists of Model 16 software and documentation and a Model 16 processor board and memory board that is installed in the card slots of any Model II. The disk drives and black and white monitor stay the same but the computing power of the Model 16, including multitasking, are made available to the Model II owner. The only limitation to this upgrade is that the maximum memory is limited to 256K.

****22**** The actual boards used in the kit and the Model 16 look like this. The lower board is the processor board and the largest IC on that board is the MC68000. The other board is one of the memory boards.

****23**** I would like to give you a quick technical look inside the Model 16. The MC68000 microprocessor is unique

among 16 bit processors in that it is the only processor with a full 32 bit internal architecture. This feature gives it a substantial speed advantage over its competition. We choose it because it is the most powerful microprocessor available.

The memory IC's in the Model 16 are 64K RAMs. The Z-80A processor is used as an input/output processor so the 68000 can handle number crunching and memory while the Z-80A handles I/O. The design includes hardware multitasking memory management for speed and efficiency and the two serial ports provide the terminal connections without need of additional hardware.

One of those serial ports is also designed for both BiSynch and SNA/SDLC communications to mainframes without the need of additional hardware. This is an important feature for large company users that is not available on most other desktop microcomputers.

****24**** This is a photo of the new thinline disk drives used in the Model 16. In addition to their half size and half weight they are the only 8 inch drives with DC motors which greatly reduces their heat output. The drives

are double sided but are capable of reading from the single sided diskettes currently used by the Model II.

Having snowed you with technical details I think it is time to consider the most important element of any computer, the customers. The buyers of today's microcomputers are not motivated by the technical features of the product, they are motivated by the applications of the computer. And the applications of these computers are totally determined by the available software. So lets look at our software plans for the Model 16 while we look at it's market.

****25**** The first market we see for the Model 16 is the small business. We have seen our sales grow as we have evolved from the Model I disk system with it's very limited accounting abilities up to the Model II, and recently up to the Model II with hard disk. But we know that there are many, many small businesses whose needs today require more capacity than a Model II and who require assurance that their needs tomorrow will not find them with an obsolete product. The Model 16 with it's multitasking ability is the product for that customer.

The software for that customer is our integrated small business accounting system. This system consists of the 6

modules shown which all share the same common data base. These are very high quality COBOL programs which were all originally implemented on 16 bit minicomputers. We have available now for the Model II all of these except payroll as floppy disk programs. They will be recompiled to operate on the Model 16 with slight modifications for multitasking. By mid-year the entire set will be in stock, and they will run on either floppy disk or hard disk.

The advantage to this accounting software is that it is fully interactive. The order entry system automatically updates the inventory and the accounts receivable which in turn updates the general ledger. The sales analysis program can prepare reports from either the accounts receivable or the order entry system. While the accounting system will operate on floppy disks it is more efficient and faster to use the hard disk. With the Model 16 the customer can grow the computer by just adding low cost terminals as his business grows and continue to operate the same software. With three terminals one could be dedicated to order entry, one to accounts receivable and the other could run the less frequently used programs like payroll, general ledger, and accounts payable.

PERSONAL COMPUTER PRICE COMPARISON

<u>COMPUTER</u>	<u>PROCESSOR</u>	<u>RAM MEMORY</u>	<u>MAX RAM MEMORY</u>	<u>DISK CAPACITY</u>	<u>DISK SIZE</u>	<u>SYSTEM PRICE</u>
IBM P.C.	8/16 Bit	128K	256K	320K	5-1/4"	\$4455
APPLE III	8 Bit	128K	256K	286K	5-1/4"	4535
TRS-80 MODEL II	8 Bit	64K	64K	1,000K	8"	5049
TRS-80 MODEL 16	16/32 Bit	128K	512K	2,500K	8"	5798
HP 125	8 Bit	64K	64K	540K	5-1/4"	6250
DEC DECMATE	12 Bit	64K	64K	1,200K	8"	7495
IBM DATAMASTER	8 Bit	64K	128K	2,200K	8"	7800

SYSTEMS INCLUDE: CPU, 2 disk drives, keyboard,
monitor, 2 RS-232 ports,
1 parallel port